

Machine Learning for Natural Language Understanding

Exercise WS 2024/2025

Mandatory Exercise

Due: 27.02.2022

Multiple Choice**10 Points**

Are the following statements true or false?

	Statement	True	False
1	The sigmoid-function $h_{\theta}(x)$ is smooth and symmetric at $x = 0$.		
2	BERT is a language model based on RNNs		
3	A Multilayer Perceptron can only represent simple linear functions		
4	Every continuous function can be approximated arbitrarily closely by a multi-layer Artificial Neural Network.		
5	GPT-3 is known for its language generation abilities		
6	RNNs capture long-term dependencies		
7	Hyperparameters should be tuned on the validation set		
8	An overfitted model performs well on unknown data		
9	The classification of unbalanced data is measured best with accuracy		
10	The harmonic mean of precision and recall is called F-measure		

Aspects of Machine Learning Models**10 Points**

- 1) What is the term for the process of converting real-world data into representations that can be used, for example, as input for neural networks? Provide an example! (2 Points)

- 2) Explain what the training, validation and test data sets are used for in machine learning. (3 Points)

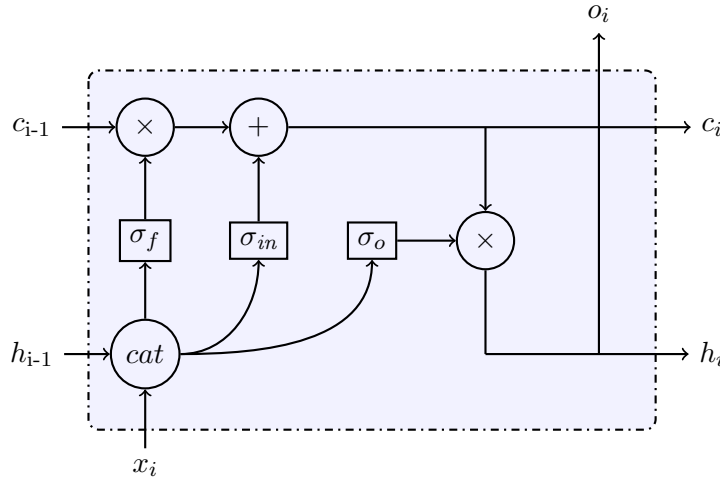
- 3) What is the difference between *supervised*, *unsupervised* and *reinforcement* learning? (3 Points)

- 4) Explain the difference between regression and classification? (2 Points)

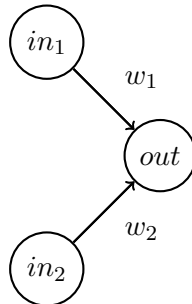
Neural Networks I

10 Points

1) Consider this simplified LSTM cell:



Assume all three gates (σ_x) to be simple feed-forward networks with two inputs and one output:



Assume the Bias to be 0. *Cat* stands for the concatenation operation in the following order: $(cat(h_{i-1}, x_i))$. **Example:** $x_1 = 5, x_2 = 3 : cat(x_1, x_2) \rightarrow [1, 5]$.

$\times$ and $+$ are multiplication and addition. Assume the following activation function for all σ :

$$f(x) = \begin{cases} 2, & \text{if } x \geq 2 \\ 1, & \text{if } x < 2 \end{cases}$$

Given the following weights and inputs, calculate the values for o_i , c_i and h_i (6 points)!

INPUTS:

$$h_{i-1} = 2$$

$$c_{i-1} = 1$$

$$x_i = 1.5$$

WEIGHTS:

$$w_{f1} = 0, w_{f2} = 0.6$$

$$w_{in1} = -0.5, w_{in2} = 0.5$$

$$w_{o1} = 0.5, w_{o2} = 0.25$$

2) Explain how the LSTM architecture prevents Gradients from vanishing. (2 Points)

3) Explain the main disadvantage of LSTMs when compared to the Transformer architecture. (In terms of scalability) (2 Points)

Neural Networks II**10 Points**

- 1)** Name 5 commonly used activation functions! Indicate the respective value ranges. *(5 Points)*

- 2)** What is the *Vanishing Gradient Problem*? *(2 Points)*

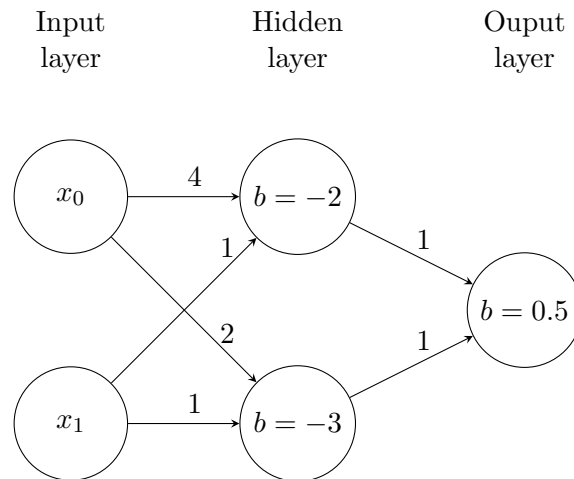
- 3)** Why is it hard to find a fitting learning rate for machine learning model training? What are the possible outcomes of a badly chosen learning rate and how can this problem be tackled? *(3 Points)*

Neural Networks III**10 Points**

1) The following table contains predicted values from a simplified linear model ($Y_i = \text{Beta}_0 + x_i$) and their true (i.e. expected) counterpart. Show the calculation of the MSE for this model! Which value should Beta_0 have to minimize the MSE? (*5 Points*)

Predicted	Expected
0	1
1	2
3	4
4	5

2) Take a look at the following Neural Network:



Calculate the network predictions and compare them with the expected classes. Assume *act* as activation function. (Apply the bias before the activation) (5 Points)

x_0	x_1	Class
1	1	1
2	1	1
-3	2	-1

$$\text{act}(x) := \begin{cases} +1, & \text{if } x > 0, \\ -1, & \text{if } x \leq 0. \end{cases}$$

